GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Diploma in Engineering

Level: Diploma

Branch: Plastics Engineering

Course Code : DI02023011

Course Name : Polymer Chemistry

w. e. f. Academic Year:	2024-25
Semester:	2nd
Category of the Course:	Professional Core Course 02

Prerequisite:	**Basic Polymer Chemistry**
Rationale:	The course deals with structures, properties & fundamentals of polymer preparation by various polymerization techniques. This course will help students to understand micro structure analysis of polymers, its solutions & degradation. Learning of this course would make students aware about the concepts of polymer chemistry and morphology for understanding the structure and manufacturing of polymers.

Course Outcome:

After Completion of the Course, Student will able to:

No	Course Outcomes	RBT Level
01	Select suitable polymerization technique.	A
02	Describe morphology and its effect on polymer.	U
03	Compare various polymer degradation mechanisms.	U
04	Identify suitable polymer reaction for polymer modification.	U
05	Select manufacturing process for a polymer.	A

**Revised Bloom's Taxonomy (RBT)*

Teaching and Examination Scheme:

Teaching Scheme (in Hours)			Total Credits L+T+ (PR/2)	Assessment Pattern and Marks				Total Marks
				Theory		Tutorial / Practical		
L	T	PR	C	ESE (E)	PA / CA (M)	PA/CA (I)	ESE (V)	
3	0	2	4	70	30	20	30	150

GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Diploma in Engineering

Level: Diploma

Branch: Plastic Engineering (Sandwich Pattern)

Course Code : DI02023011

Course Name : Polymer Chemistry

Course Content:

Unit No.	Content	No. of Hours	% of Weightage
UNIT – I	**Polymerization Technique** - Introduction - Bulk Polymerization - Solution Polymerization - Suspension Polymerization - Emulsion Polymerization	06	15% (10 Marks)
UNIT– II	**Glass Transition Temperature & Polymer Structure** - Introduction • Glass transition temperature and associated properties • Factors affecting Tg • Significance of Tg - Polymer morphology - Polydispersity - Degree of polymerization - Molecular weight of polymer • Concept • Molecular weight distribution • Significance of molecular weight - Microstructure based on (Chemical structure & Geometric structure) - Polymer crystallinity • Concept • Number average & weight average molecular weight • Crystalline and amorphous structure of polymer • Crystallisability • Factors affecting crystallisability • Effect of crystallinity on polymer properties	12	25% (20 Marks)
UNIT– III	**Polymer degradation** - Introduction - Types of degradation (Chain-end & Random) - Degradations – factors & its impact and remedies of: • Thermal • Mechanical	06	15% (10 Marks)

	• Ultrasonic wave • Photo • Oxidative • Hydrolytic		
UNIT – IV	**Polymer Reaction and Polymer Solution** - Introduction - Hydrolysis - Acidolysis - Aminolysis - Hydrogenation - Addition reactions - Substitution reactions - Cross-linked reactions	06	15% (10 Marks)
UNIT – V	**Industrial Manufacturing of Polymer** - Introduction - Industrial anufacturing process of polymers: • Low density polyethylene (LDPE) • High density polyethylene (HDPE) • Poly Propylene (PP) • Poly Vinyl Chloride (PVC) • Poly Styrene (PS) • Acrylo Nitryl Butadine Styrene (ABS) • Poly Amide (Nylon) • Poly Carbonate • Poly Methyl Methacrylate (PMMA) • Phenol Formaldehyde (PF) • Urea Formaldehyde (UF) • Melamine Formaldehyde (MF)	12	30% (20 Marks)
	Total	**42**	**100% (70 Marks)**

Suggested Specification Table with Marks (Theory):

Distribution of Theory Marks (in %)					
R Level	**U Level**	**A Level**	**N Level**	**E Level**	**C Level**
35%	**35%**	**30%**	**-**	**-**	**-**

Where R: Remember; U: Understanding; A: Application, N: Analyze and E: Evaluate C: Create (as per Revised Bloom's Taxonomy)

References/Suggested Learning Resources:

(a) Books:

S. No.	Title of Book	Author	Publication with place, year and ISBN
1	Textbook of Polymer Science	Fred W. Billmeyer	John Wiley & sons, Singapore,2009, 978-0-471- 03196-3
2	Polymer Science	Govariker V.R	New Age International Pub, Delhi, 2019, 9788122438130
3	Polymer Science and Technology	Fried J.R	Prentice Hall, Delhi, 2014, 9780137039555
4	Polymer Science and Technology	Ghosh Pramamoy	Tata McGrow Hill Education Pvt. Ltd, Delhi, 2010, 9780070707047
5	Polymer Chemistry	Charles E. Carraher Jr.	CRC Press, Delhi, 2017, 9781498737388
6	Plastics Materials	J A Brydson	ISBN: 978-0-7506-4132-6 Publisher: Elsevier Science Published: 22 Nov 1999
7	Polymer Degradation and Stabilization	W. L. Hawkins	ISBN :9783642693762 Publisher: Springer Berlin Heidelberg Published:6 December 2012
8	Thermoplastic Materials: Properties, Manufacturing Methods, and Applications	Christopher C. Ibeh	ISBN 1420093835 Publisher CRC Press Published 25 April 2011

(b) Open source software and website:

a) **https://ndl.iitkgp.ac.in**

b) **https://onlinecourses.nptel.ac.in**

c) **https://swayam.gov.in/explorer**

d) **www.cheresources.com**

Suggested Course Practical List:

Sr. No.	Practical Outcomes (PrOs)	Unit No.	Approx. Hrs. required
1	Compare different types of polymerization techniques with respect to its physical parameters.	1	02
2	Prepare Polystyrene by bulk polymerization technique.	1	04
3	Determine T_g of High Density Polyethylene by Dilatometer method.	2	02
4	Determine number average molecular weight of Polypropylene by End Group Analysis.	2	02
5	Determine molecular weight of polystyrene using Ostwald viscometer.	2	02
6	Compare different types of polymer degradation method.	3	02
7	Prepare vinyl alcohol from PVAc using polymer modification method.	3	04
8	Prepare MMA monomer from depolymerization of PMMA.	3	04
9	Prepare PMMA using suspension polymerization technique.	5	04
10	Manufacture Polycaprolactum by interfacial condensation polymerization technique.	5	04
11	Prepare polyester from ethylene glycol and maleic anhydride by interfacial polymerization.	5	04
12	Prepare Phenol Formaldehyde by polycondensation process.	5	04
		Total	**38**

List of Laboratory/Learning Resources Required:

Sr. No.	Equipment Name with Broad Specifications
1	Dilatometer
2	Ostwald viscometer with fittings
3	Various glasswares
4	Laboratory scale polymerizer set-up
5	Drying oven

Suggested Project List:

1. Prepare comparative chart for different polymerization techniques
2. Prepare a chart of polymers showing classification based on microstructure
3. Compare different polymer degradations
4. Prepare charts for various polymer reactions
5. Prepare model of polymerizer
6. Prepare a flow chart of manufacturing a polymer

Suggested Activities for Students:

1. Assignments
2. Technical Quiz/MCQ Test
3. Presentation on some course topic
4. I-net based assignments
5. Students are encouraged to register themselves in various MOOCS such as: Swayam, edx, Coursera, Udemy etc to further enhance their learning

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